Project Title

(Type your project title here)

Project Report submitted for the partial fulfilment of grade for the course

Data Structures and Algorithms -3 (25CS2103E)

in

A.Y. 2026-2027

Submitted by
Niyathi Konduparthy (2520090072)
Chakrika Reddy (2520090044)

Team Number: 23

Section: 10

Under the guidance of
Ms. K. Chadusha

Asst. Professor, CSIT



Hyderabad-500090, Telangana, India.

K L (Deemed to be) University

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING / COMPUTER SCIENCE AND INFORMATION TECHNOLOGY/ARTIFICIAL INTELLIGENCE & DATA SCIENCE



DECLARATION

The Mini Project Report entitled "**Community Detection Platform"** is a record of Bonafide work of **Niyathi Konduparthy – 2520090072, Chakrika Reddy– 2520090044,** submitted in partial fulfillment for the award of B. Tech in Computer Science and Engineering/Computer Science and Information Technology/ Artifical Intelligence & data Science to the K L University. The results embodied in this report have not been copied from any other departments/University/Institute.

Niyathi Konduparthy – 2520090072

Chakrika Reddy – 2520090044

K L (Deemed to be) University

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING / COMPUTER SCIENCE AND INFORMATION TECHNOLOGY



CERTIFICATE

This is to certify that the mini project based report entitled "**Community Detection Platform"** is a bonafide work done and submitted by **Niyathi Konduparthy – 2520090072, Chakrika Reddy – 2520090044** in partial fulfillment of the requirements for the award of the degree of **BACHELOR OF TECHNOLOGY** in Department of Computer Science Engineering/ Computer Science and Information Technology/Artificial Intelligence & Data Science, K L (Deemed to be University), during the academic year **2026-2027.**

Signature of the Supervisor

Signature of the HOD **Signature of the External Examiner**

ACKNOWLEDGEMENT

The success in this project would not have been possible but for the timely help and guidance rendered by many people. Our wish to express my sincere thanks to all those who has assisted us in one way or the other for the completion of my project.

Our greatest appreciation to my guide, <<Guide Name>>, <<Designation>>, <Department>> which cannot be expressed in words for his/her tremendous support, encouragement and guidance for this project.

We express our gratitude to **Dr. P.Venkateshwar Rao (CSE)/Dr. Khaja Shareef (CSIT)/ Dr. Sreenath Itikela (AI&DS)**, Head of the *Department for Computer Science Engineering/ Department for Computer Science and Information Technology/Department of Artificial Intelligence and Data Science* for providing us with adequate facilities, ways and means by which we are able to complete this project-based Lab.

We thank all the members of teaching and non-teaching staff members, and also who have assisted me directly or indirectly for successful completion of this project. Finally, I sincerely thank my parents, friends and classmates for their kind help and cooperation during my work.

Niyathi Konduparthy– 2520090072

Chakrika Reddy – 2520090044

TABLE OF CONTENTS

S.No.	Contents	Page no.
1.	Abstract	
2.	Introduction	
3.	Literature survey	
4.	Problem Statement and Objectives	
5.	Hardware and Software requirements	
6.	Implementation	
7.	Results & Discussion	
8.	Conclusion & Future Scope	
9.	References	

1.ABSTRACT

The Community-Detection Platform is a graph-based analytical system designed to identify groups of closely connected users within large social, communication, or collaboration networks. The platform constructs a weighted interaction graph, where each node represents a user or entity and each edge represents the strength of interaction between two users. Edge weights can be determined using factors such as frequency of communication, common connections, and similarity in shared content.

Using graph-based community detection algorithms, the system partitions the network into meaningful communities whose members are more strongly connected to one another than to the rest of the network. The platform can support and compare multiple community detection approaches, allowing users to analyse how different algorithms perform on the same dataset. It also evaluates the quality of detected communities using graph metrics such as modularity, density, conductance, community size, and connectivity.

An interactive visualization module can display the network structure and highlight different communities, making complex relationships easier to understand. Users can explore individual nodes, connections, community statistics, and interaction patterns. The system may also include filtering and search features to analyse specific users or communities efficiently.

The platform emphasizes the practical application of graph theory, data structures, algorithms, and performance analysis. It can be applied in social network analysis, recommendation systems, targeted digital marketing, collaboration analysis, fraud detection, and cybersecurity. Overall, the project provides an efficient and scalable approach for discovering hidden structures and meaningful relationships within complex network data.

Team Name: 23

Team Members:

1. Niyathi Konduparthy, 2520090072
2. Chakrika Reddy, 2520090044

2. INTRODUCTION

The **Community Detection Platform** is a data-driven application developed using Data Structures and Algorithms (DSA) to identify groups of users or entities that share similar interests, behaviors, or interaction patterns.

The platform represents users and their interactions as a **weighted graph**, where nodes represent users or entities and edges represent connections between them. The weights of these edges indicate the strength of relationships based on factors such as interactions, shared interests, and content similarity.

By applying graph-based algorithms and network analysis techniques, the platform identifies communities within the network and evaluates their structure using metrics such as modularity, density, and conductance. It also provides visual representations of the detected communities, helping users understand relationships and patterns within complex networks.

The primary objective of this project is to demonstrate how fundamental DSA concepts, particularly graphs, graph traversal, and network analysis, can be applied to solve real-world problems involving community identification and relationship analysis.

3. LITERATURE SURVEY

Community detection is an established research area in graph theory and network analysis. It focuses on identifying groups of closely connected nodes within complex networks. Existing research has explored various algorithms and techniques for discovering communities in social networks, biological networks, and other interconnected systems.

1. Community Detection in Networks: A Multidisciplinary Review (2018)

Javed et al. reviewed community detection methods and their applications across different domains. The study discussed traditional graph-based techniques, overlapping community detection, and challenges in analyzing complex networks. It established the importance of selecting suitable algorithms based on the network structure and application requirements.

2. Community Detection in Node-Attributed Social Networks: A Survey (2020)

Petr Chunaev examined community detection approaches that consider both network connections and node attributes. The study highlighted the limitations of methods that rely only on graph structure, showing the importance of incorporating additional information about users, such as their interests and characteristics.

3. A Comprehensive Review of Community Detection in Graphs (2024)

This study examined multiple categories of community detection methods, including modularity-based approaches, spectral clustering, probabilistic models, and deep-learning techniques. It also discussed the evaluation and comparison of community detection algorithms on different graph datasets.

4. A Survey of Community Detection Algorithms and Its Comparative Performance Analysis (2025)

Singh and Garg reviewed community detection research published between 2002 and 2024. Their work discussed algorithms such as Girvan–Newman, Label Propagation, Infomap, and Louvain, along with methods for evaluating the quality of detected communities.

Research Gap

Existing research provides many algorithms for identifying communities in graphs. However, there remains a practical need for accessible platforms that combine relationship modelling, weighted graph analysis, community identification, evaluation metrics, and understandable visualizations in a unified interface.

Our project addresses this practical requirement by integrating graph-based community analysis and visualization into a single platform. It aims to make community structures easier to explore and interpret while demonstrating the application of DSA concepts to real-world network problems.

.

4. Problem Statement and Objectives

Problem Statement: To develop a graph-based platform that identifies meaningful communities by analyzing relationships and interaction strengths between users.

Problem Description: The platform represents users as nodes and their relationships as weighted edges to detect closely connected communities. It evaluates community structures using graph-based metrics and visualizes the results for easy interpretation.

Objectives:

1. To model relationships using graphs: Represent users or entities as nodes and their relationships as weighted edges.
2. To implement graph-based algorithms: Apply suitable Data Structures and Algorithms techniques to analyze network connections and identify communities.
3. To incorporate relationship strength: Use edge weights to represent the strength of interactions or similarity between connected users.
4. To identify meaningful communities: Group closely connected users or entities based on the structure and weights of the network.

5. HARDWARE AND SOFTWARE REQUIREMENTS

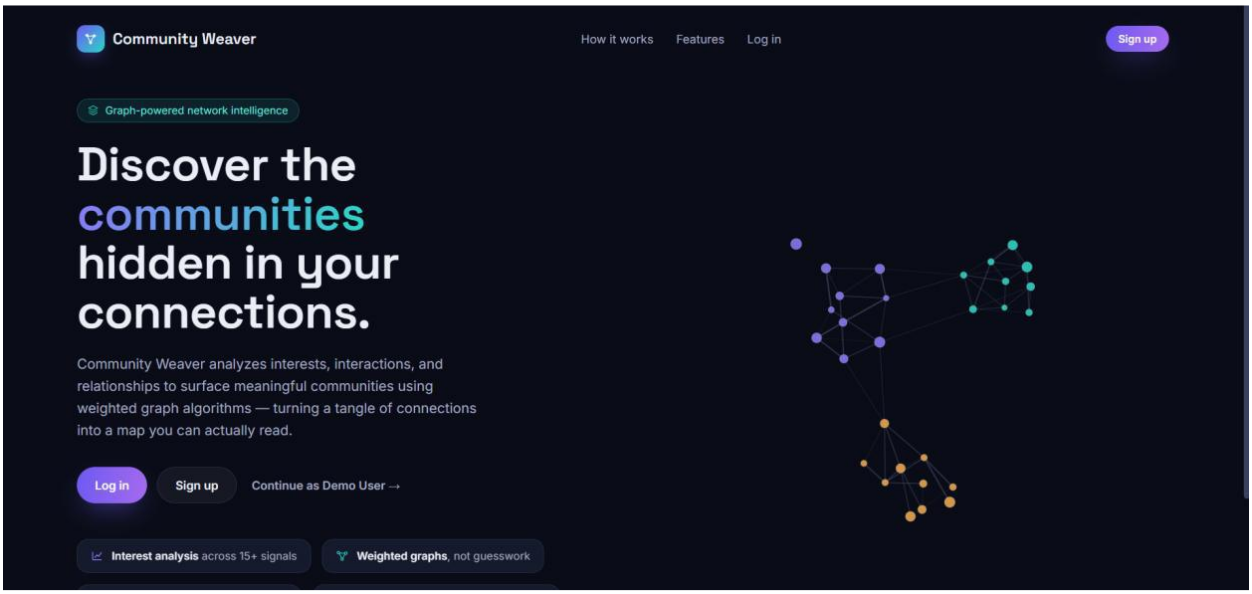
Hardware Requirements:

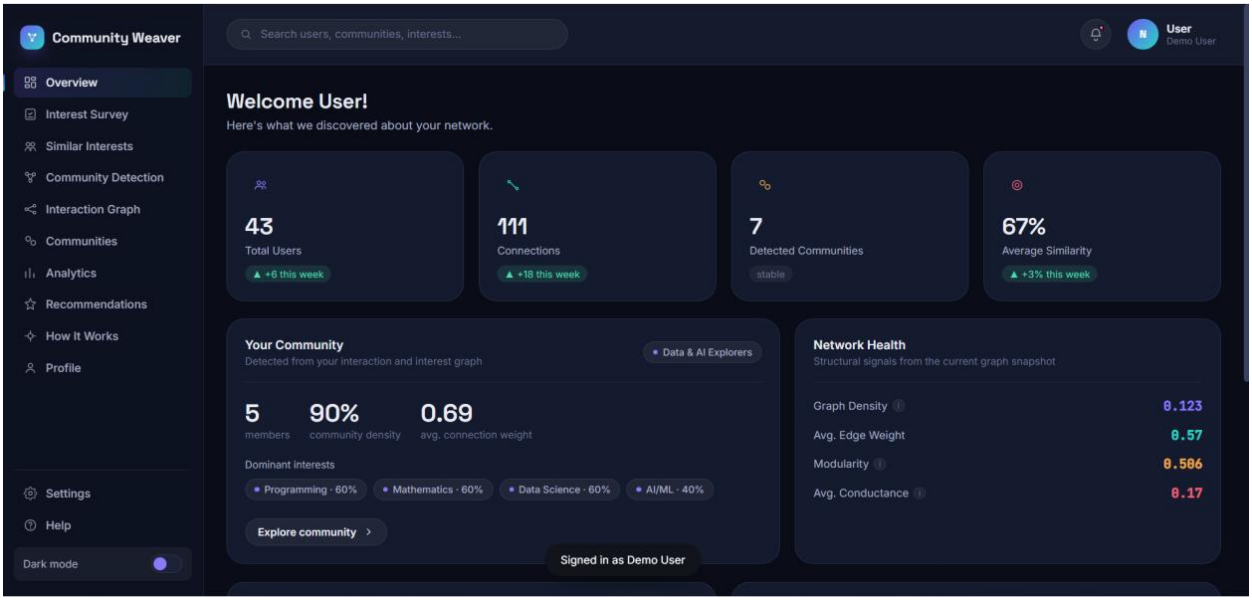
- i5 / Ryzen 5 processor
- 8 GB RAM (16 GB recommended)
- GPU optional; internet required for setup

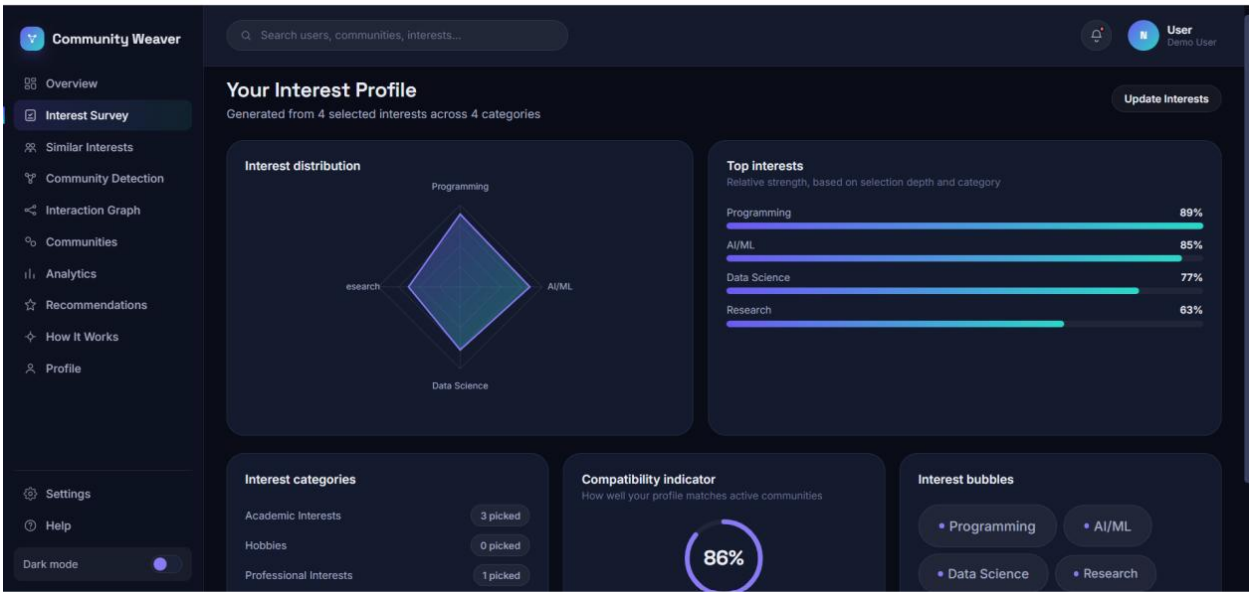
Software Requirements:

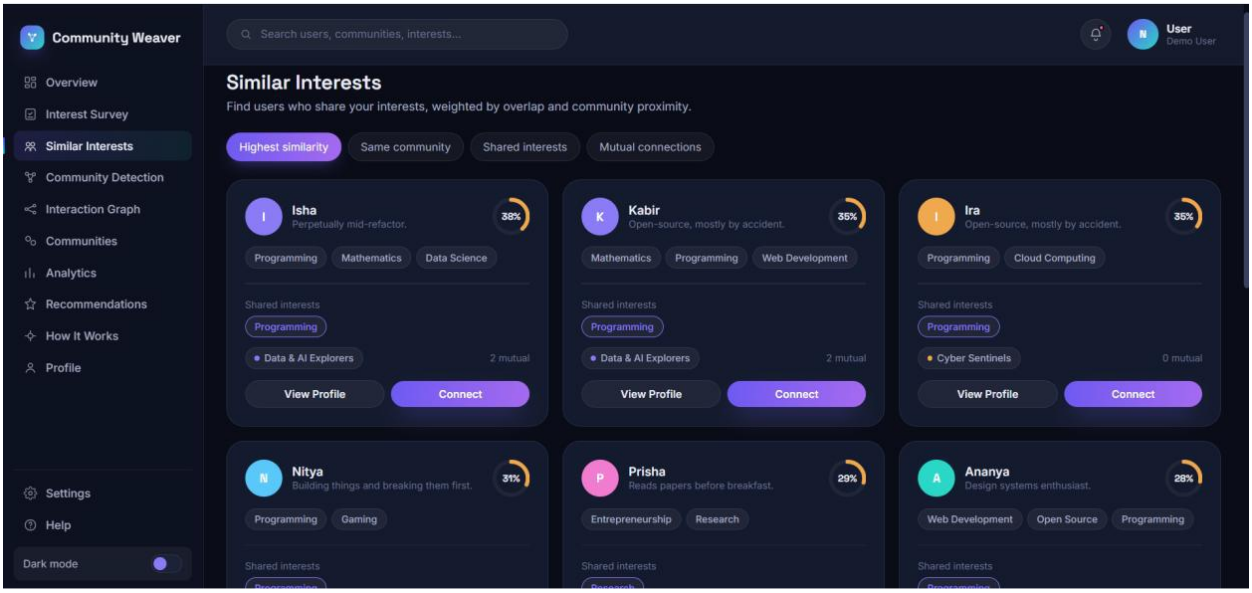
- Python 3.x
- NetworkX
- Pandas
- Flask

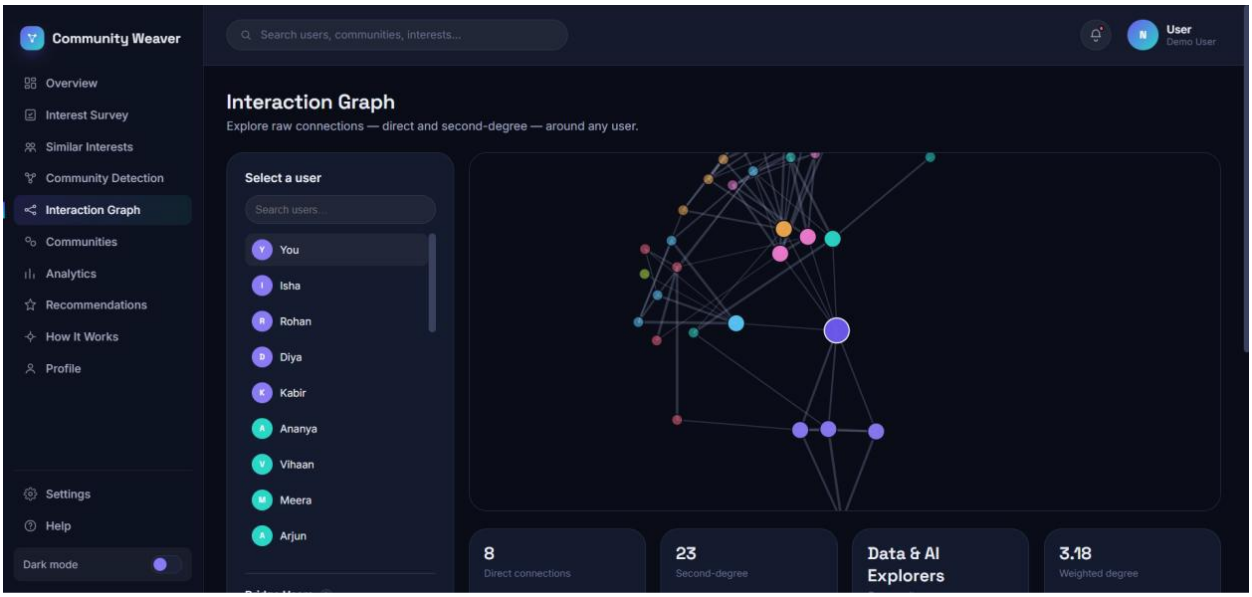
5. IMPLEMENTATION






Community Weaver
Search users, communities, interests...
User
Demo User
Overview
Interest Survey
Similar Interests
Community Detection
Interaction Graph
Communities
Analytics
Recommendations
How It Works
Profile
Settings
Help
Dark mode
Your Interest Profile
Generated from 4 selected interests across 4 categories
Update Interests
Interest distribution
Programming
AI/ML
Data Science
Top interests
Relative strength, based on selection depth and category
Programming 89%
AI/ML 85%
Data Science 77%
Research 63%
Interest categories
Academic Interests 3 picked
Hobbies 0 picked
Professional Interests 1 picked
Compatibility indicator
How well your profile matches active communities
86%
Interest bubbles
Programming
AI/ML
Data Science
Research


Community Weaver
Search users, communities, interests...
User
Demo User
Overview
Interest Survey
Similar Interests
Community Detection
Interaction Graph
Communities
Analytics
Recommendations
How It Works
Profile
Settings
Help
Dark mode
Similar Interests
Find users who share your interests, weighted by overlap and community proximity.
Highest similarity
Same community
Shared interests
Mutual connections
Isha
Perpetually mid-refactor.
38%
Programming
Mathematics
Data Science
Shared interests
Programming
Data & AI Explorers
2 mutual
View Profile
Connect
Kabir
Open-source, mostly by accident.
35%
Mathematics
Programming
Web Development
Shared interests
Programming
Data & AI Explorers
2 mutual
View Profile
Connect
Ira
Open-source, mostly by accident.
35%
Programming
Cloud Computing
Shared interests
Programming
Cyber Sentinels
0 mutual
View Profile
Connect
Nitya
Building things and breaking them first.
31%
Programming
Gaming
Shared interests
Prisha
Reads papers before breakfast.
29%
Entrepreneurship
Research
Shared interests
Ananya
Design systems enthusiast.
28%
Web Development
Open Source
Programming
Shared interests


Community Weaver
Search users, communities, interests...
User
Demo User
Overview
Interest Survey
Similar Interests
Community Detection
Interaction Graph
Communities
Analytics
Recommendations
How It Works
Profile
Settings
Help
Dark mode
Interaction Graph
Explore raw connections — direct and second-degree — around any user.
Select a user
Search users...
You
Isha
Rohan
Diya
Kabir
Ananya
Vihaan
Meera
Arjun
8
Direct connections
23
Second-degree
Data & AI Explorers
3.18
Weighted degree

6. RESULTS & DISCUSSION

Results

The Community Detection Platform successfully demonstrates the representation of user relationships using a weighted graph, where nodes represent users and edges represent the strength of their connections. The platform identifies groups of closely connected users and provides a visual representation of the network. Community evaluation metrics, including modularity, density, and conductance, can be used to analyze the structure and quality of the detected communities.

Discussion

The results demonstrate how graph-based Data Structures and Algorithms can be applied to identify meaningful patterns in interconnected networks. Weighted relationships help represent connection strengths, while graph visualization improves the understanding of community structures. The effectiveness of community detection depends on the quality of the input data, the selected algorithm, and the choice of evaluation metrics. The platform can be further enhanced by testing larger real-world datasets and comparing different community detection algorithms.

7. CONCLUSION & FUTURE SCOPE

The Community Detection Platform demonstrates the practical application of Data Structures and Algorithms, particularly weighted graphs, to identify meaningful communities within interconnected networks. By analyzing relationships and visualizing community structures, the platform simplifies network analysis and provides insights into user connections. This project establishes a foundation for efficient community identification and further graph-based analysis.

- **Real-Time Analysis:** Detect communities dynamically as new users and interactions are added.
- **Advanced Algorithms:** Integrate and compare algorithms such as Louvain, Label Propagation, and Girvan–Newman.
- **Larger Datasets:** Support large-scale real-world social networks and interaction datasets.
- **Improved Visualization:** Develop interactive graph visualizations with community highlighting and relationship exploration.
- **AI Integration:** Explore machine learning techniques to improve community identification using user interests and behavioral patterns.
- **Practical Applications:** Extend the platform to social networks, recommendation systems, research collaborations, and online interest groups.

.

8. REFERENCES

1. Newman, M. E. J. (2006). Modularity and community structure in networks. *Proceedings of the National Academy of Sciences*, 103(23), 8577–8582. https://doi.org/10.1073/pnas.0601602103

2. Blondel, V. D., Guillaume, J.-L., Lambiotte, R., & Lefebvre, E. (2008). Fast unfolding of communities in large networks. *Journal of Statistical Mechanics: Theory and Experiment*, 2008(10), P10008. https://doi.org/10.1088/1742-5468/2008/10/P10008

3. Fortunato, S. (2010). Community detection in graphs. *Physics Reports*, 486(3–5), 75–174. https://doi.org/10.1016/j.physrep.2009.11.002

4. Traag, V. A., Waltman, L., & van Eck, N. J. (2019). From Louvain to Leiden: guaranteeing well-connected communities. *Scientific Reports*, 9, 5233. https://doi.org/10.1038/s41598-019-41695-z

5. Javed, M. A., Younis, M. S., Latif, S., Qadir, J., & Baig, A. (2018). Community detection in networks: A multidisciplinary review. *Journal of Network and Computer Applications*, 108, 87–111. https://doi.org/10.1016/j.jnca.2018.02.011

6. NetworkX Developers. *NetworkX Documentation: Community Detection Algorithms and Modularity*. https://networkx.org/documentation/stable/reference/algorithms/community.html

7. NetworkX Developers. *NetworkX Documentation: Modularity*. https://networkx.org/documentation/stable/reference/algorithms/generated/networkx.algorithms.community.quality.modularity.html